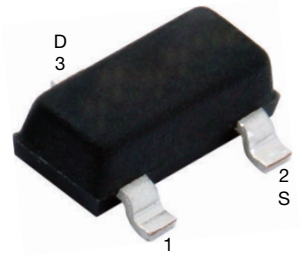
Automotive P-Channel 40 V (D-S) 175 °C MOSFET

SOT-23 (TO-236)



Top View

Marking Code: 9UYXX

FEATURES

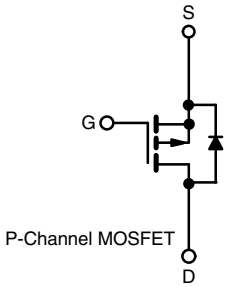
- TrenchFET® power MOSFET
- AEC-Q101 qualified
- 100 % R_g and UIS tested
- Material categorization:
  for definitions of compliance please see www.vishay.com/doc?99912

AUTOMOTIVE GRADE

Pb-free

RoHS COMPLIANT

HALOGEN FREE



PRODUCT SUMMARY	
V_{DS} (V)	-40
$R_{DS(on)}$ (Ω) at V_{GS} = -10 V	0.094
$R_{DS(on)}$ (Ω) at V_{GS} = -4.5 V	0.188
I_D (A)	-4.1
Configuration	Single

ORDERING INFORMATION	
Package	SOT-23
Lead (Pb)-free and halogen-free	SQ2389CES (for detailed order number please see www.vishay.com/doc?79771)

ABSOLUTE MAXIMUM RATINGS (T_C = 25 °C, unless otherwise noted)

PARAMETER		SYMBOL	LIMIT	UNIT
Drain-source voltage		V_{DS}	-40	V
Gate-source voltage		V_{GS}	± 20	
Continuous drain current	T_C = 25 °C	I_D	-4.1	A
	T_C = 125 °C		-2.4	
Continuous source current (diode conduction)		I_S	-3.6	
Pulsed drain current [a]		I_{DM}	-16	
Single pulse avalanche current	L = 0.1 mH	I_{AS}	-12	
Single pulse avalanche energy		E_{AS}	7.2	mJ
Maximum power dissipation	T_C = 25 °C	P_D	3	W
	T_C = 125 °C		1	
Operating junction and storage temperature range		T_J, T_{stg}	-55 to +175	°C

THERMAL RESISTANCE RATINGS

PARAMETER		SYMBOL	LIMIT	UNIT
Junction-to-ambient	PCB mount [b]	R_{thJA}	166	°C/W
Junction-to-foot (drain)		R_{thJF}	50	

Notes

a. Pulse test; pulse width ≤ 300 µs, duty cycle ≤ 2 %

b. When mounted on 1" square PCB (FR4 material)

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SPECIFICATIONS (T_C = 25 °C, unless otherwise noted)							
PARAMETER	**SYMBOL**	**TEST CONDITIONS**		**MIN.**	**TYP.**	**MAX.**	**UNIT**
Static							
Drain-source breakdown voltage	V_{DS}	V_{GS} = 0 V, I_D = -250 μA		-40	-	-	V
Gate-source threshold voltage	$V_{GS(th)}$	V_{DS} = V_{GS}, I_D = -250 μA		-1.5	-2.0	-2.5	
Gate-source leakage	I_{GSS}	V_{DS} = 0 V, V_{GS} = ± 20 V		-	-	± 100	nA
Zero gate voltage drain current	I_{DSS}	V_{GS} = 0 V	V_{DS} = -40 V	-	-	-1	μA
		V_{GS} = 0 V	V_{DS} = -40 V, T_J = 125 °C	-	-	-50	
		V_{GS} = 0 V	V_{DS} = -40 V, T_J = 175 °C	-	-	-150	
On-state drain current [a]	$I_{D(on)}$	V_{GS} = -10 V	V_{DS} ≤ -5 V	-10	-	-	A
Drain-source on-state resistance [a]	$R_{DS(on)}$	V_{GS} = -10 V	I_D = -3 A	-	0.084	0.094	Ω
		V_{GS} = -10 V	I_D = -3 A, T_J = 125 °C	-	-	0.144	
		V_{GS} = -10 V	I_D = -3 A, T_J = 175 °C	-	-	0.169	
		V_{GS} = -4.5 V	I_D = -3 A	-	0.140	0.188	
Forward transconductance [b]	g_{fs}	V_{DS} = -10 V, I_D = -3 A		-	5	-	S
Dynamic [b]							
Input capacitance	C_{iss}	V_{GS} = 0 V	V_{DS} = -20 V, f = 1 MHz	-	414	455	pF
Output capacitance	C_{oss}			-	80	100	
Reverse transfer capacitance	C_{rss}			-	42	54	
Total gate charge [c]	Q_g	V_{GS} = -10 V	V_{DS} = -20 V, I_D = -3 A	-	8.2	12	nC
Gate-source charge [c]	Q_{gs}			-	1.1	-	
Gate-drain charge [c]	Q_{gd}			-	3.0	-	
Gate resistance	R_g	f = 1 MHz		3.1	4.1	7	Ω
Turn-on delay time [c]	$t_{d(on)}$	V_{DD} = -20 V, R_L = 6.7 Ω I_D ≅ -3 A, V_{GEN} = -10 V, R_g = 1 Ω		-	7	10	ns
Rise time [c]	t_r			-	12	16	
Turn-off delay time [c]	$t_{d(off)}$			-	16	20	
Fall time [c]	t_f			-	4	8	
Source-Drain Diode Ratings and Characteristics [b]							
Pulsed current [a]	I_{SM}			-	-	-10	A
Forward voltage	V_{SD}	I_F = -1.5 A, V_{GS} = 0 V		-	-0.8	-1.2	V
Body diode reverse recovery time	t_{rr}	I_F = -2 A, di/dt = 100 A/μs		-	17	34	ns
Body diode reverse recovery charge	Q_{rr}			-	14	28	nC
Reverse recovery fall time	t_a			-	14	-	ns
Reverse recovery rise time	t_b			-	3	-	
Body diode peak reverse recovery current	$I_{RM(REC)}$			-	-1.99	-	A

Notes

a. Pulse test; pulse width ≤ 300 μs, duty cycle ≤ 2 %
b. Guaranteed by design, not subject to production testing
c. Independent of operating temperature

Stresses beyond those listed under "Absolute Maximum Ratings" may cause permanent damage to the device. These are stress ratings only, and functional operation of the device at these or any other conditions beyond those indicated in the operational sections of the specifications is not implied. Exposure to absolute maximum rating conditions for extended periods may affect device reliability.

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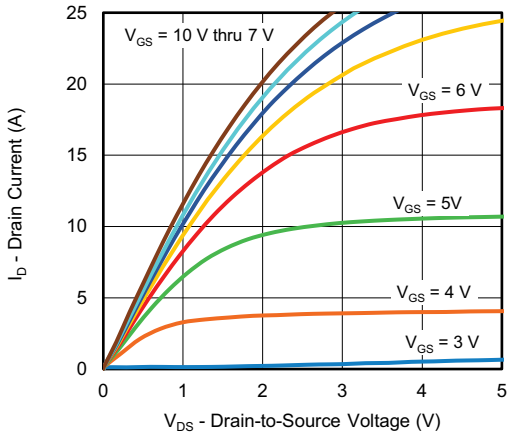
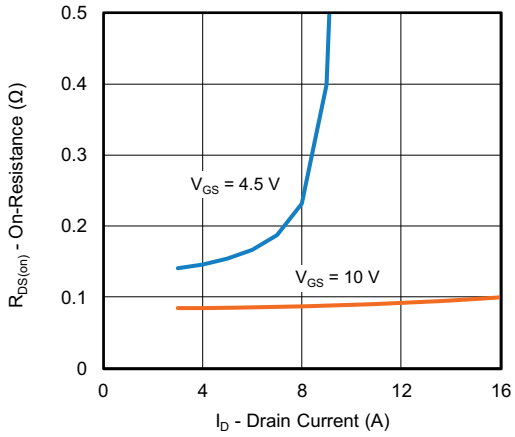
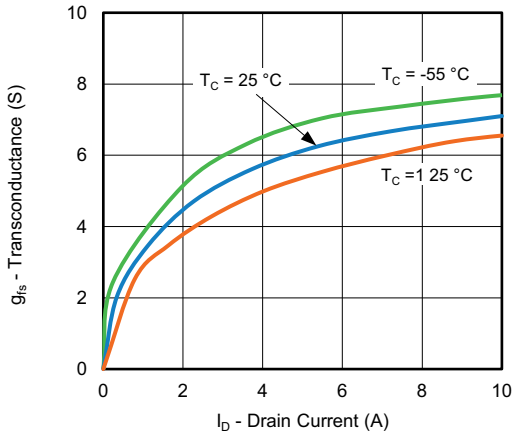
TYPICAL CHARACTERISTICS (T_A = 25 °C, unless otherwise noted)



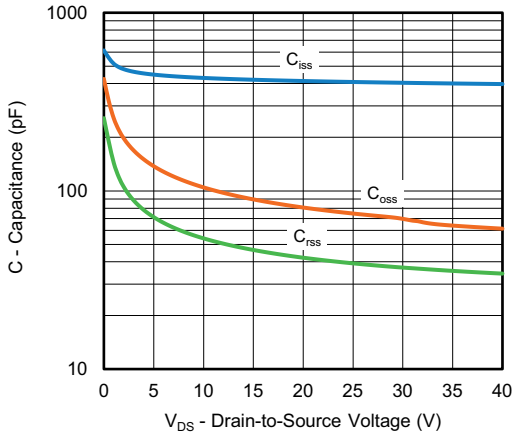
Output Characteristics



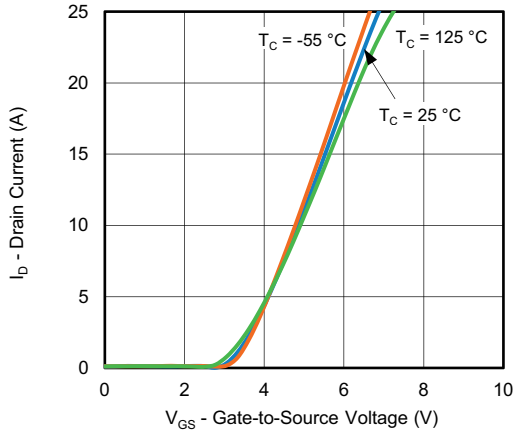
On-Resistance vs. Drain Current



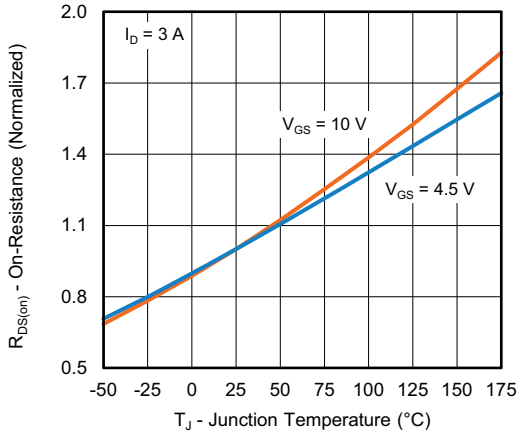
Transconductance



Capacitance



Transfer Characteristics



On-Resistance vs. Junction Temperature

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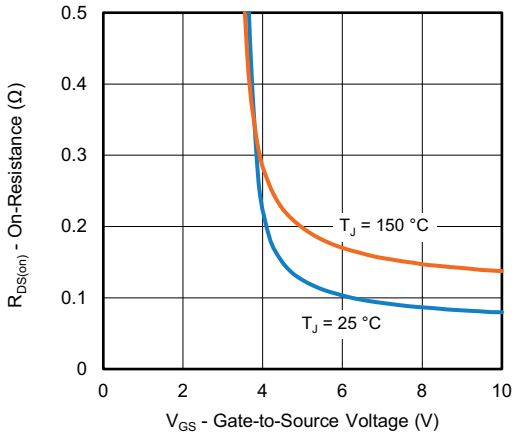
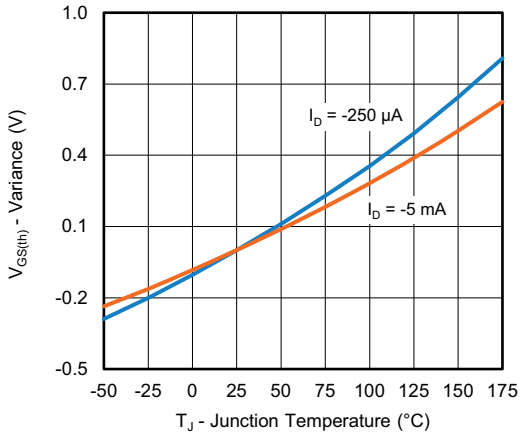
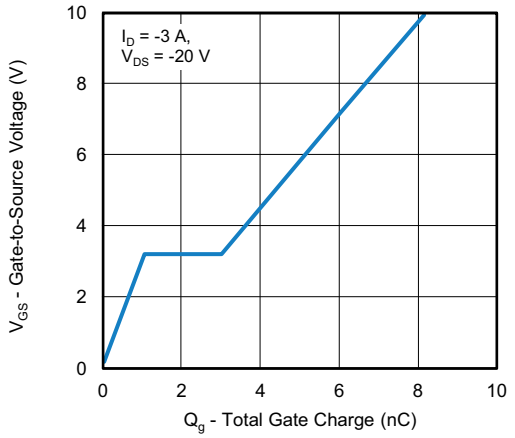
TYPICAL CHARACTERISTICS (T_A = 25 °C, unless otherwise noted)



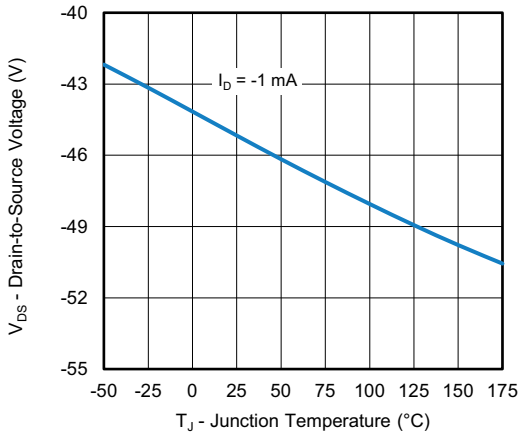
On-Resistance vs. Gate-Source Voltage



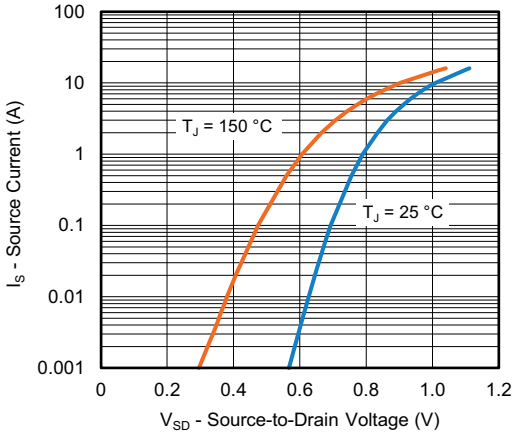
Threshold Voltage



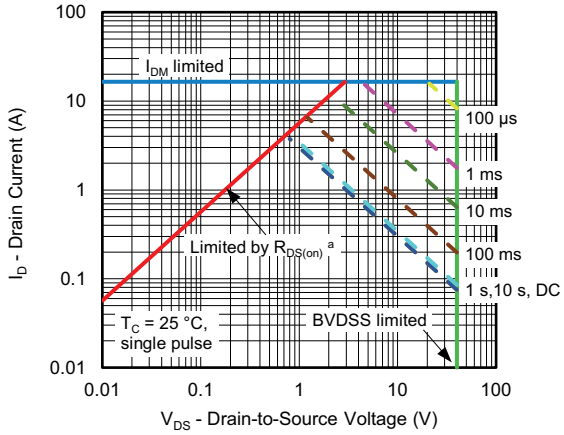
Gate Charge



Drain Source Breakdown vs. Junction Temperature



Source-Drain Diode Forward Voltage



Safe Operating Area

Note

a. V_{GS} > minimum V_{GS} at which $R_{DS(on)}$ is specified

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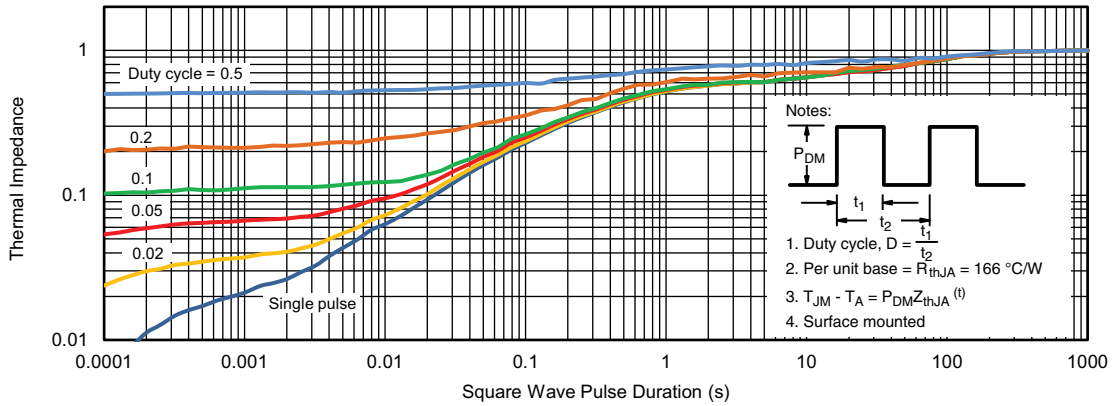
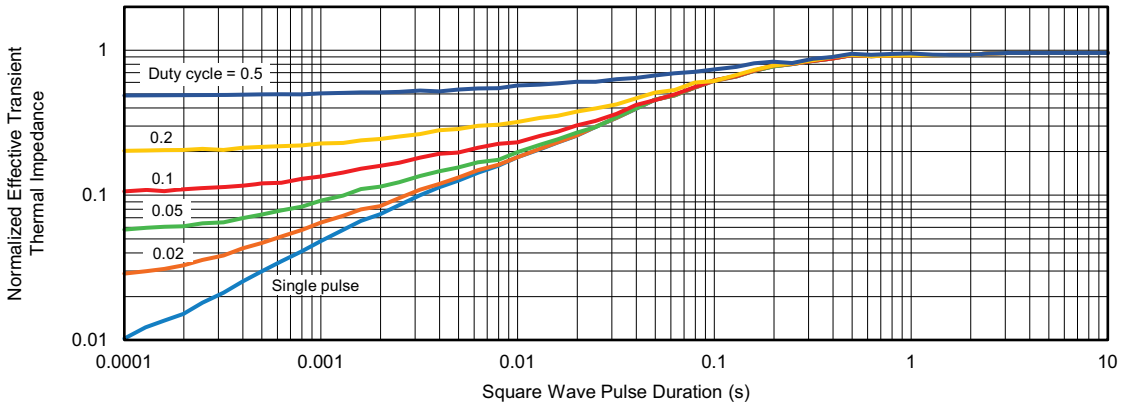
THERMAL RATINGS (T_A = 25 °C, unless otherwise noted)



Normalized Thermal Transient Impedance, Junction-to-Ambient



Normalized Thermal Transient Impedance, Junction-to-Foot

Notes

- The characteristics shown in the two graphs
  - Normalized Transient Thermal Impedance Junction-to-Ambient (25 °C)
  - Normalized Transient Thermal Impedance Junction-to-Foot (25 °C)

  are given for general guidelines only to enable the user to get a “ball park” indication of part capabilities. The data are extracted from single pulse transient thermal impedance characteristics which are developed from empirical measurements. The latter is valid for the part mounted on printed circuit board - FR4, size 1" x 1" x 0.062", double sided with 2 oz. copper, 100 % on both sides. The part capabilities can widely vary depending on actual application parameters and operating conditions

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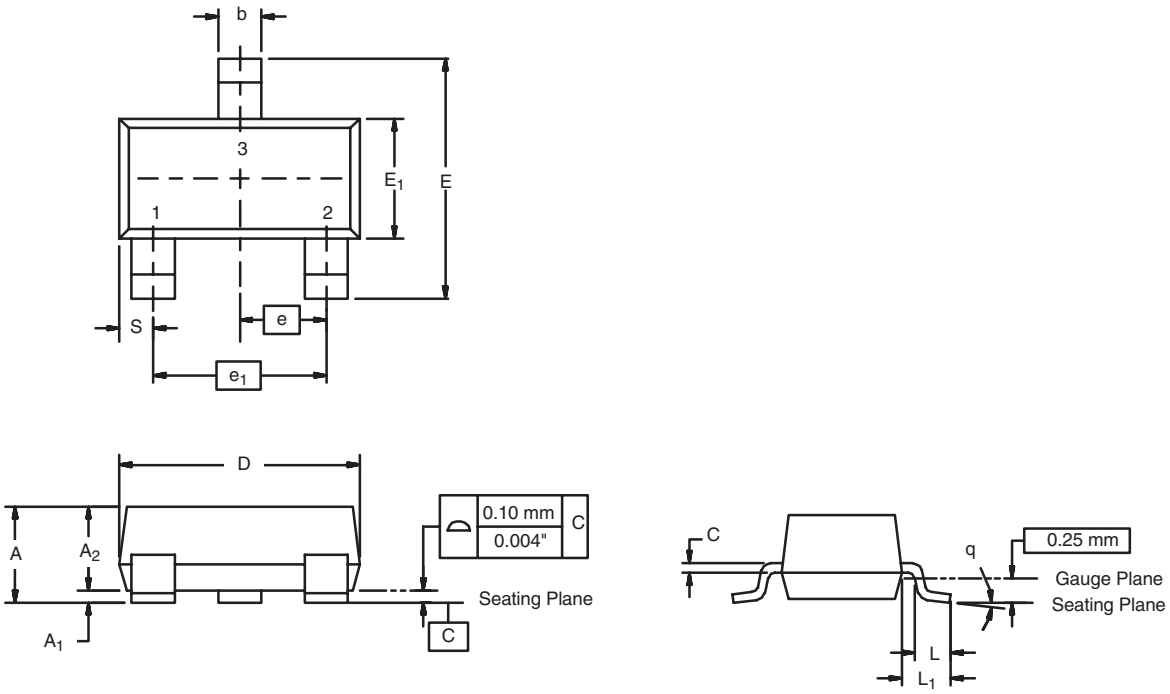
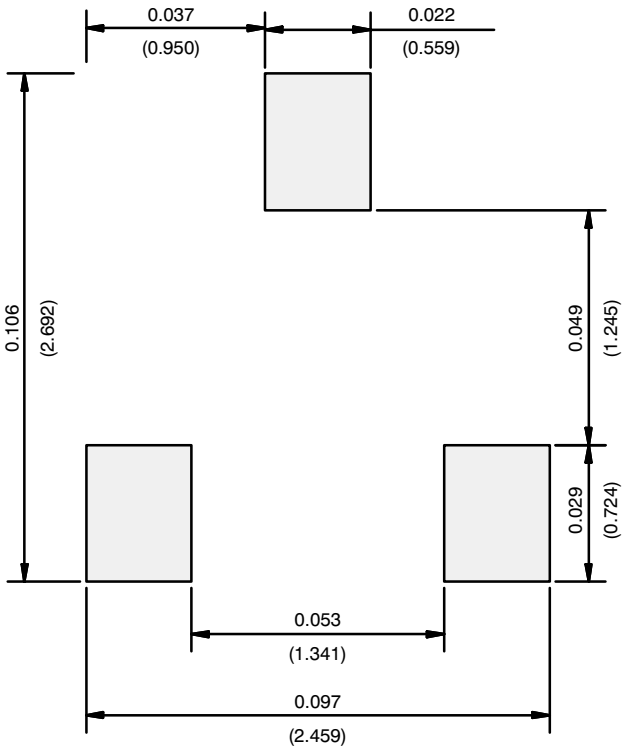
SOT-23 (TO-236): 3-LEAD



Dim	MILLIMETERS		INCHES	
	Min	Max	Min	Max
A	0.89	1.12	0.035	0.044
A_1	0.01	0.10	0.0004	0.004
A_2	0.88	1.02	0.0346	0.040
b	0.35	0.50	0.014	0.020
c	0.085	0.18	0.003	0.007
D	2.80	3.04	0.110	0.120
E	2.10	2.64	0.083	0.104
E_1	1.20	1.40	0.047	0.055
e	0.95 BSC		0.0374 Ref	
e_1	1.90 BSC		0.0748 Ref	
L	0.40	0.60	0.016	0.024
L_1	0.64 Ref		0.025 Ref	
S	0.50 Ref		0.020 Ref	
q	3°	8°	3°	8°
ECN: S-03946-Rev. K, 09-Jul-01 DWG: 5479				

RECOMMENDED MINIMUM PADS FOR SOT-23



Recommended Minimum Pads
Dimensions in Inches/(mm)

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